

Research Paper Report Manual

1. Find a Suitable Research Paper

Choose **one recent, credible research paper** that:

- Matches your domain/topic.
- Uses a real dataset, not toy/simulated data.
- Uses understandable methods such as TF-IDF, K-Means, DBSCAN, Sentiment Analysis, N-gram, Apriori, Correlation, etc.
- Has a publisher/DOI/official link.
- Has enough experiment details: dataset, baselines, metrics, and results.

2. Understand the Paper

Read and extract only the necessary parts:

- Abstract
- Introduction
- Method / Algorithm section
- Dataset / Experimental Setup
- Results / Evaluation section

From these, write:

- Problem
- Dataset
- Main method
- Input → Process → Output pipeline
- Evaluation metrics
- Main result

Golden rule: if the paper does not clearly say something, write **“not clearly stated in provided text.”**

3. Find One Research Gap

Do not choose many gaps. Select **one small, practical gap**.

Possible gap types:

- Missing metric
- Missing simple baseline
- Hyperparameter not tested
- Missing ablation / component test

For the selected gap, write:

- Gap statement
- Evidence from the paper
- Why this gap matters
- Why it is feasible for a student project

4. Plan the Mini-Extension

Based on the selected gap, propose a small extension.

Write:

- Extension title
- Exact change you will make
- Dataset to use
- Method / algorithm to use
- Metrics for comparison
- Expected output: table, graph, confusion matrix, screenshot, etc.

Keep it realistic: laptop or free Colab, no huge model training, no unnecessary new data collection.

5. Implement or Verify

Now do the practical work.

Minimum work:

- Reproduce the original method as much as possible.
- Apply your small change.
- Compare original vs your extension.
- Save results in tables/figures.

If you use substitute data or default parameters, clearly write:

ASSUMPTION: We used this because the original paper did not clearly provide it.

Final Report Structure

1. Introduction

Write:

- Topic background
- Problem statement
- Why the paper is important
- What your project tries to analyze or improve

2. Background & Original Paper Summary

Clearly label this as **Original Paper**.

Include:

- Paper title, authors, year
- Main problem
- Dataset
- Main method
- Main result

3. Method Understanding

Explain the original paper's pipeline:

Input → Preprocessing → Algorithm/Model → Output

Also mention:

- Features used
- Hyperparameters if stated
- Evaluation metrics

4. Original Experiments Summary

Create a small table:

Item	Details
Dataset	
Baseline	
Method	
Metric	
Key Result	

This section should explain what was tested, on which dataset, against which baseline, and using which metric.

5. Gap Analysis

Write:

- Selected gap
- Evidence from the paper
- Why it is a valid gap
- Why it is feasible to test

6. Proposed Extension

Clearly label this as **Our Extension**.

Include:

- What you changed
- Why you changed it
- What result you expected
- What metric you used to check it

7. Implementation & Results

Include:

- Tools/libraries used
- Dataset used
- Steps followed
- Result table
- Graphs/screenshots if available

Compare: Original Paper vs Our Extension

8. Discussion

Write:

- Did your extension improve anything?
- If yes, why?
- If no, why not?
- What limitations remain?
- What assumptions did you make?

9. Conclusion

Write:

- What the original paper did
- What gap you found
- What extension you tried
- What you learned from the result

10. Quality Checklist

Before submission, check:

- No fake paper facts.
- Original Paper and Our Extension are clearly separated.
- Every missing detail is marked as “not clearly stated.”
- The gap is evidence-based.
- The extension is small and realistic.
- Results are shown using table/figure.
- Assumptions are clearly labeled.

Shortest workflow:

Find paper → Understand method → Find one gap → Plan small extension → Implement/verify → Write report with “Original Paper” and “Our Extension” separated.